

JIVESH ARORA

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EDUCATION

University of Waterloo, Canada

Jan 2025 - Present

Bachelor of Honours Computer Science, Co-op

Relevant Coursework: Data Structures and Algorithms, Statistics for CS, Databases, Logic and Computation

Research: Incoming URA under Prof. Marina Meila (manifold learning), Fall 2026

PROJECTS

cold-start-recsys | [Project](#) | [Source Code](#)

Tech Stack: Python, PyTorch, Sentence-Transformers, FastAPI, Docker, Streamlit, AWS SageMaker

- Built a hybrid recommender combining PyTorch matrix factorization with sentence-transformer embeddings for cold-start users/items on 800K+ Amazon Reviews, lifting Recall@10 by 1.17pp via confidence-weighted fusion.
- Fixed a lookahead leak and a scale-mismatch bug in fusion scoring via leave-one-out validation.
- Deployed as a containerized FastAPI service on a live AWS SageMaker endpoint with a Streamlit demo UI.

Lost and Found Web App | [Website](#) | [Source Code](#)

Tech Stack: MongoDB, Express.js, React, Node.js, Nodemailer, Railway, Vercel

- Developed full-stack MERN app with JWT authentication, serving 100+ users at 99.5% uptime.
- Architected RESTful API handling 500+ calls; reduced item recovery time by 60%.

OCC Community Chatbot | [Project](#) | [Source Code](#)

Tech Stack: Node.js, Express.js, Groq API (Llama 3.3 70B), SQLite, React

- Built a 4-stage agent pipeline (router → retrieval → action agent → critic) serving 5,000+ students.
- Implemented autonomous function-calling for ticket creation, follow-ups, and escalation via a bounded tool loop.
- Designed a rule-based critic to override unsafe outputs and force-escalate high-risk cases, logged for eval.

EXPERIENCE

Royal Bank Of Canada (RBC)

January 2026 - August 2026

Insight Analyst, Digital Analytics (Personal & Commercial Banking)

Toronto, Canada

- Built an automated PySpark → Hive → Tableau ETL pipeline for national campaign reporting, replacing manual refresh workflows with DAG/YAML-based Airflow scheduling to cut reporting turnaround time.
- Diagnosed a Tableau fan-out bug inflating an iPad campaign metric ~27x; fixed with FIXED LOD expressions.
- Analyzed provincial-level response rates for RBC's national iPad marketing campaign against spend data, surfacing regional performance gaps that directly informed reallocation of marketing budget.
- Migrated legacy QuickSight reporting to Tableau, converting SQL logic to PySpark and pushing to Git.

Off-Campus Community, WUSA, University of Waterloo

August 2025 - April 2026

Off-Campus Don

Waterloo, Canada

- Support 5,000+ off-campus students with housing inquiries, resources, and community guidance.
- Organize social events and activities to foster community engagement and inclusion among off-campus students.

Statistics Club, University of Waterloo

April 2025 - December 2025

Events Lead

Waterloo, Canada

- Organize 10+ workshops per term, growing attendance 45%; manage \$2,000+ budgets.

SKILLS

Certifications

AI/ML

Data & Pipelines

Languages

Backend & Cloud

Tools

AWS Certified Cloud Practitioner (CLF-C02), AWS Certified AI Practitioner (AIF-C01)

PyTorch, Sentence-Transformers, NLP, LLMs (OpenAI, Groq), RAG, FAISS, LangChain

PySpark, Hadoop, Hive, Spark, SQL, Airflow, DAG/YAML scheduling, Tableau

Python, JavaScript, TypeScript, C, C++, SQL, R

Node.js, Express.js, Flask, FastAPI, AWS, Docker, RESTful APIs, MongoDB, PostgreSQL

Git/GitHub, Postman, Linux, Vercel, Railway